

Mathematics Diagnostic Baseline Assessment for Grade IX students

Based on Grade VIII Mathematics

Time: 30 minutes | Total Marks: 25

Instructions

Choose the one best answer for each question. Attempt all questions. Each question carries 1 mark. You may use rough paper for working.

Domain A. Number & Operations and Sets

1. Which pair contains one rational number and one irrational number?

A. $\sqrt{25}$ and 0.4

B. $\sqrt{8}$ and π

C. 0.121212 ... and $\frac{7}{8}$

D. $\sqrt{49}$ and $\sqrt{10}$

2. The number

$$5.18181818 \dots$$

is best described as:

A. a rational terminating decimal

B. a rational non-terminating recurring decimal

C. an irrational number

D. an integer

3. What is

$$\sqrt{0.0081} ?$$

A. 0.9

B. 0.009

C. 0.09

D. 0.081

4. Eight workers can complete a job in 15 days. At the same rate, how many days would 12 workers take?

- A. 10 days
- B. 12 days
- C. 18 days
- D. 22.5 days

5. A shopkeeper marks an item at Rs 4,000 and gives a 10% discount. The item originally cost the shopkeeper Rs 3,000.

What percentage profit does the shopkeeper make on the **cost price**?

- A. 15%
- B. $16\frac{2}{3}\%$
- C. 20%
- D. $33\frac{1}{3}\%$

6. A set B has 4 elements. How many elements are in its power set $P(B)$?

- A. 8
- B. 4
- C. 32
- D. 16

7. Which statement is always true?

- A. The product of two irrational numbers is always irrational.
 - B. The square root of every positive integer is irrational.
 - C. Every non-terminating decimal is irrational.
 - D. The sum of a rational number and an irrational number is always irrational.
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8. Solve:

$$-3x + 4 > 10$$

A. $x < -2$

B. $x > -2$

C. $x < 2$

D. $x > 2$

9. Simplify:

$$3x(2x - 5) - 2x(x - 4)$$

A. $4x^2 - 23x$

B. $8x^2 - 7x$

C. $4x^2 - 7x$

D. $4x^2 + 7x$

10. Which expression is equal to

$$(3x - 2)^2?$$

A. $9x^2 - 4$

B. $9x^2 - 12x + 4$

C. $9x^2 - 6x + 4$

D. $6x^2 - 12x + 4$

11. Which is the correct factorization of

$$x^2 + x - 12?$$

A. $(x - 4)(x + 3)$

B. $(x + 6)(x - 2)$

C. $(x + 12)(x - 1)$

D. $(x + 4)(x - 3)$

12. Which ordered pair satisfies both equations?

$$x + y = 9$$

$$x - y = 3$$

A. (6, 3)

B. (3, 6)

C. (7, 2)

D. (5, 4)

13. Simplify:

$$\frac{(a^3b^2)(a^2b^4)}{ab^3}$$

A. a^4b^6

B. a^5b^3

C. a^4b^3

D. a^6b^9

14. What is the degree of the polynomial $4x^3y^2 - 7xy^4 + 3$?

A. 5

B. 2

C. 4

D. 8

15. The arithmetic sequence 7, 12, 17, 22, ... continues in the same way. Which expression gives its n th term?

A. $5n + 7$

B. $7n - 2$

C. $5n + 2$

D. $5n - 2$

16. A straight line passes through the points $(-1, 2)$ and $(3, 10)$. What is the gradient of the line?

- A. $\frac{1}{2}$
- B. 4
- C. -2
- D. 2

Domain C. Measurement & Geometry

17. A ladder 15 m long is placed against a wall. If its upper end reaches the height of 12 m along the wall, then find the distance of the foot of the ladder from the wall.

- A. 3 m
- B. 6 m
- C. 9 m
- D. 12 m

18. A cone has a base area of 30 cm^2 and a perpendicular height of 9 cm.

What is its volume?

- A. 90 cm^3
- B. 270 cm^3
- C. 30 cm^3
- D. 810 cm^3

19. Choose the correct statement.

- A. If two parallel lines are intersected by a transversal, the interior angles on the same side of the transversal are congruent.
- B. If two parallel lines are intersected by a transversal, the interior angles on the same side of the transversal are supplementary.
- C. The diagonals of every rectangle bisect each other at right angles.
- D. If a transversal intersects any two lines, the alternate interior angles are always congruent.

20. A circle has radius 7 cm. An arc subtends an angle of 90° at the centre.

Using $\pi = \frac{22}{7}$, what is the length of the arc?

- A. 5.5 cm
- B. 22 cm
- C. 44 cm
- D. 11 cm

21. In a right-angled triangle, if the opposite side = 4 cm and the hypotenuse = 5 cm, then $\sin \theta = ?$

- A. $\frac{4}{5}$
- B. $\frac{3}{5}$
- C. $\frac{5}{4}$
- D. $\frac{1}{2}$

22. What is the sum of the interior angles of a 7-sided polygon?

- A. 720°
 - B. 900°
 - C. 1080°
 - D. 1260°
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D. Statistics & Probability

23. Find the median of:

12, 4, 7, 2, 5

- A. 6
 - B. 5
 - C. 7
 - D. 10
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24. A fair coin is tossed and a fair six-sided die is rolled.

What is the probability of getting a head and an even number?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{3}{4}$

D. $\frac{1}{6}$

25. The scores 2, 4, 6 and 8 occur with frequencies 1, 3, 4 and 2 respectively. What is the mean score?

A. 5.0

B. 5.4

C. 6.0

D. 5.6

Mathematics Diagnostic Baseline Assessment - Answer Key

Grade IX | Based on Grade VIII Mathematics | 25 Marks

Question	Answer	Diagnostic Concept
1	D	Rational and irrational numbers
2	B	Recurring decimals
3	C	Square roots of decimals
4	A	Inverse proportion
5	C	Financial arithmetic / profit percentage
6	D	Power set
7	D	Properties of rational and irrational numbers
8	A	Linear inequality
9	C	Simplification of algebraic expressions
10	B	Algebraic identity
11	D	Factorization
12	A	Simultaneous linear equations
13	C	Laws of indices
14	A	Degree of a polynomial
15	C	Arithmetic sequence / nth term
16	D	Gradient of a straight line
17	C	Pythagoras theorem
18	A	Volume of a cone
19	B	Parallel lines and transversal
20	D	Arc length
21	A	Trigonometric ratio (sine)
22	B	Interior angles of polygons
23	B	Median
24	A	Combined probability
25	B	Mean from a frequency distribution